Polymer Chemistry

ROYAL SOCIETY OF CHEMISTRY

COMMUNICATION

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Polym. Chem.*, 2020, **11**, 5578

Received 16th July 2020,
Accepted 25th August 2020

DOI: 10.1039/d0py01018a

rsc.li/polymers

Published on 26 August 2020. Downloaded by Fudan University on 2/20/2021 3:08:06 AM.

Vinyl-addition polymerizations of cycloallenes: synthetic access to congeners of cyclic-olefin polymers†

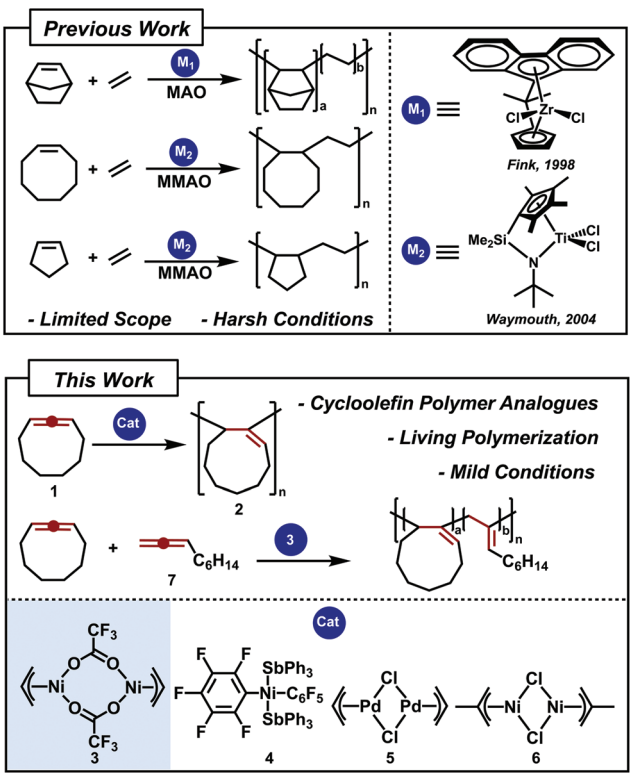
Nicholas J. Galan,‡ Justin M. Burroughs,‡ Christopher R. Maroon, Brian K. Long and Johnathan N. Brantley *

Expanding the scope of cycloolefin-type monomers that polymerize *via* coordination–insertion mechanisms is an enduring synthetic challenge. Vinyl-addition polymerizations of cyclic olefins are almost exclusively limited to norbornene scaffolds and/or copolymerizations with ethylene. To circumvent these synthetic limitations, we report the vinyl-addition polymerization of cycloallenes using the catalyst, [(π-allyl)NiOCOCF$_3$]$_2$. The polycyclic materials could be accessed in excellent yields (up to 97%), and the polymerization exhibited living character (which was leveraged for copolymerizations with acyclic monomers). Importantly, we found that multiple ring sizes could be incorporated into polymer architectures, and material properties could be tailored by controlling the cycloallene feed ratios.

Relative to their acyclic congeners, cycloolefin polymers are enticing synthetic targets that frequently exhibit unique physical properties. Indeed, the incorporation of small carbocyclic motifs into soft materials can impact optical transparency,[1,2] modify viscosity,[3] increase T_g,[4] and enhance the modulus.[5] In principle, cyclic motifs can be introduced into polymer matrices through a variety of polymerization manifolds; however, practical considerations often narrow the scope of applicable methods. For example, radical and ionic polymerizations can undergo deleterious chain termination events or undesired rearrangements (resulting in low yields and oligomeric products).[6,7] Furthermore, ring-opening polymerizations typically require highly strained monomers (*e.g.* norbornene), which intrinsically limits monomer scope.

Vinyl-addition polymerizations of cyclic olefins typically exhibit improved performance, but such polymerizations are not without synthetic obstacles. For example, vinyl-addition homopolymerizations of cyclic olefins other than norbornene and cyclopentene are virtually unknown.[8] Although many cyclic olefins can be copolymerized with ethylene, this strategy usually requires highly reactive catalysts (which can exhibit poor functional group tolerance) and harsh co-catalysts or activators (Scheme 1). Given these limitations, exploring new monomer and catalyst space is imperative for overcoming these obstacles.

Much like their olefin congeners, allenes polymerize under a variety of conditions[9] (*e.g.* radical,[10] cationic,[11] anionic,[12] and coordination–insertion[13–16]). We were particularly inspired by the work of Endo and colleagues, who have shown that



Scheme 1 Routes to cycloolefin polymers.

University of Tennessee Department of Chemistry, 1420 Circle Dr, Knoxville, TN 37996-1600, USA. E-mail: jbrant13@utk.edu

†Electronic supplementary information (ESI) available. See DOI: 10.1039/d0py01018a

‡These authors contributed equally.

This journal is © The Royal Society of Chemistry 2020

Published on 26 August 2020. Downloaded by Fudan University on 2/20/2021 3:08:06 AM.

[(π-allyl)NiOCOCF$_3$]$_2$ (**3**; Scheme 1) catalyzes the living polymerization of allenes bearing an array of functional groups (*e.g.* alcohols,[17] esters,[18] or amides,[19]). One brief report by Endo and colleagues revealed that catalyst **3** can facilitate the polymerization of cycloallenes; however, this succinct note lacked detailed analyses of polymerization kinetics, living character at higher conversions, copolymerizations, or bulk material properties.[20] As these materials can be considered analogues of cycloolefin polymers, we envisioned that a more detailed study of their syntheses and properties could reveal important design elements for creating a spate of advanced architectures. Indeed, cycloolefins can be efficiently transformed into cycloallenes using the two-step Skattebøl rearrangement.[21] This operationally simple strategy, combined with a deeper understanding of cycloallene polymerization tendencies and the functional group tolerance of **3**, could significantly expand the scope of potential monomers[22] for vinyl-addition polymerizations. Given these considerations, we envisioned that vinyl-addition polymerizations of cyclic allenes would afford access to novel derivatives of cycloolefin (co) polymers.

We selected 1,2-cyclononadiene (**1**) as our model substrate, given that (i) it is easily synthesized from cyclooctene on a multi-gram scale, and (ii) the resultant polymer would contain rings that are difficult to incorporate using traditional monomers.[23] Our initial studies commenced with the homopolymerization of **1** using catalyst **3** (Scheme 1). Gratifyingly, the polymerization proceeded rapidly (*vide infra*) to afford the desired polymer, **2**, as a white powder in excellent yield (94%, Table 1). Analysis by gel-permeation chromatography (GPC) revealed that polymer **2** exhibited low dispersity ($Đ$ = 1.11) and a number-average molecular weight (M_n = 36 kDa) similar to the targeted value. We note that we consistently observed a high molecular weight impurity that we were unable to avoid.[24] Thermogravimetric analysis of **2** revealed a relatively high decomposition temperature (T_d = 350 °C), which is comparable to cycloolefin polymers.[25] Unfortunately, we were unable to observe any thermal transitions using differential scanning calorimetry.

The predicted[26] ring strain of **1** is typical for other 9-membered rings (~12 kcal mol^{-1}); thus, the rapid polymerization we observed suggested that unusually high strain may not be required to access our target materials. To benchmark this, we monitored homopolymerizations of **1** and an acyclic analogue, 1,2-nonadiene (**7**), *via* ^{1}H NMR spectroscopy (see ESI†). We observed >95% conversion of **1** within 20 min, compared to approximately 60 min for >95% consumption of **7**. Furthermore, **1** exhibited a 6-fold increase in polymerization rate relative to **7** (*cf.* k_{obs} = 0.296 for **1** *versus* k_{obs} = 0.052 for **7**; Fig. 1). These studies demonstrate that modestly-strained cycloallenes possess enhanced reactivity relative to acyclic allenes. This behaviour starkly contrasts cyclic olefin polymerization tendencies, and these promising results suggest that our strategy could accommodate monomers bearing larger ring sizes (*vide infra*).

Having identified **3** as a highly active catalyst, we sought to further optimize the polymerization. We began our studies by examining other vinyl-addition polymerization catalysts[27] (**4–6**; Scheme 1). Although **1** polymerized in the presence of catalyst **4**, GPC analysis revealed that the M_n of the isolated material was significantly higher than the theoretical value (Table 1). Polymerizations in the presence of catalyst **5** (activated with $AgSbF_6$) required lengthy reaction times (24 h), whereas catalyst **6** (also activated with $AgSbF_6$) afforded low yields of highly insoluble materials (Table 1). The addition of $AgSbF_6$ to **3** (which is known to enhance the activity of Ni based catalysts[27]) resulted in materials with low M_n and broad $Đ$. Finally, polymerizations at elevated temperatures (35, 50, and 65 °C) afforded materials with (i) M_n values that exceeded the theore-

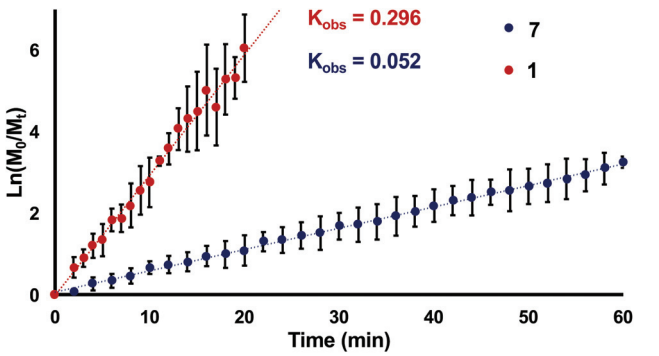


Fig. 1 Kinetic plots for polymerization of **1** (red) and **7** (blue).

Table 1 Initial catalyst screen and polymerization optimization

Entry	Catalyst[g]	$[1]_0$	Time (h)	Temperature (°C)	M_n^{theo} (Da)	M_n^{expt} [c] (Da)	$Đ$[d]	Yield (%)
1	**3**	0.1	1	23	24 400[a]	35 600	1.11	94
2	**3**	0.1	1	35	24 400[a]	50 500	1.11	93
3	**3**	0.1	1	50	24 400[a]	48 900	1.13	97
4	**3**	0.1	1	65	24 400[a]	73 300	1.12	53
5	**4**	1.5	1	23	24 400[b]	133 000	1.62	57
6	**5**[f]	1.5	24	23	24 400[a]	32 000	1.37	63
7[e]	**6**[f]	1.5	1	23	24 400[a]	—	—	59
8	**3**[f]	1.5	24	23	24 400[a]	8100	2.73	99

[a] Calculated using $[1]_0$: [**Metal**]$_0$ × 122.12. [b] Calculated using $[1]_0$: [**Catalyst**]$_0$ × 122.12. [c] Measured by GPC (triple detection). [d] Calculated using M_w/M_n. [e] Insoluble material. [f] Activated with $AgSbF_6$. [g] See Scheme 1.

Published on 26 August 2020. Downloaded by Fudan University on 2/20/2021 3:08:06 AM.

tical values (M_n = 50, 49, and 73 kDa, respectively); (ii) slightly higher dispersities ($Đ$ = 1.13, 1.12); and (iii) reduced yields, which we hypothesize may be due to catalyst decomposition (Table 1).

With our selected polymerization conditions in hand, we next evaluated the living character of these polymerizations. Briefly, aliquots were periodically removed during the polymerization of **1** and quenched by exposure to air. Monomer conversion was determined *via* ^{1}H NMR spectroscopy, and M_n and $Đ$ values were determined by GPC. As shown in Fig. 2, a linear correlation between M_n and monomer conversion was observed while maintaining relatively narrow dispersities ($Đ \leq$ 1.2). Moreover, chain extension *via* sequential addition of **1** was successful (see ESI for more details†), further supporting the living nature of this polymerization (Fig. 2). Importantly, these data suggest that cycloallenes may be readily integrated into an array of soft materials through copolymerizations.

Given that polymers containing cyclic motifs often exhibit unique thermomechanical properties, we sought to investigate the effect of cycloallene-derived moieties on mechanical properties (*i.e.* Young's modulus). In a preliminary study, we synthesized a series of high M_n (~100 kDa) statistical copolymers by varying the feed ratio of **1** : **7** (70 : 30, 50 : 50, and 30 : 70; Table 2). These materials were then cast into films from $CHCl_3$ and used for subsequent tensile testing. We surmised that increased cycloallene content would result in an increased Young's modulus (in direct analogy to reports of more traditional cycloolefin copolymers).[28] Conversely, we expected increased acyclic allene content would lower the modulus [in direct analogy to poly(α-olefins)].

Tension tests were conducted for materials containing 70 and 50 mol% **1**. The homopolymer of **1** proved too brittle for analysis, whereas both the homopolymer of **7** and the 30 mol% copolymer proved too tacky for mechanical analysis. We measured a Young's modulus of 1367 ± 15 MPa [strain at break = 2.5% (±0.6%)] for the 70 mol% polymer (Fig. 3). Conversely, tensile testing of the 50 mol% material revealed a Young's modulus of 263 ± 30 MPa [strain at break = 5.2% (±2.4%)]. These data are consistent with observations from cyclic olefin polymers,[28] and they demonstrate that relatively modest changes in cycloallene feed ratios can dramatically impact the mechanical properties of the resultant materials. Slouf reported similar trends in copolymers with varied norbornene content (15–30% norbornene),[29] which further illustrates the apparent similarity between cycloolefin and cycloallene polymers. Although preliminary, these results suggest that our approach could be amenable to synthesizing a spate of soft materials with highly tunable physical properties.

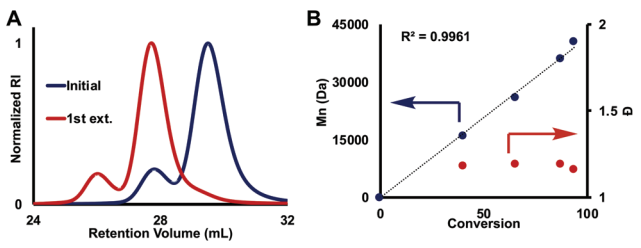


Fig. 2 (A) GPC plot showing chain extension of **1**. (B) M_n *vs.* conversion plot of **1**.

Table 2 Polymerization results for (co)polymers of **1** and **7**

Entry	(**1** : **7**)	Yield (%)	M_n [a] (Da)	$Đ$ [b]
1	100 : 0	76	61 356	1.32
2	0 : 100	74	67 570	1.15
3	70 : 30	97	129 100	1.38
4	50 : 50	92	98 360	1.85
5	30 : 70	78	99 270	1.75

[a] Measured by GPC. [b] Calculated using M_w/M_n.

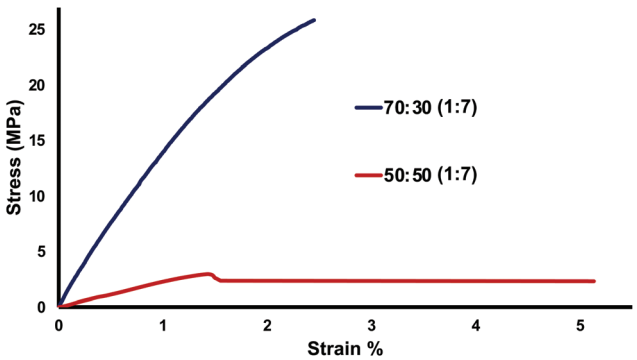


Fig. 3 Stress *vs.* strain plots for copolymers of **1** and **7**. Results for 70 : 30 (**1** : **7**, blue) and 50 : 50 (red) copolymer films are shown.

Finally, we sought to demonstrate the compatibility of our polymerization strategy with other cycloallene ring sizes. Synthesizing materials with large ring sizes is inherently difficult,[30] with cycloolefin scaffolds being limited primarily to fused polycycles (*e.g.* tetracyclododecene). However, increased ring sizes could further modify bulk material properties, such as T_d [31] or modulus.[28] We prepared 1,2-cyclotridecadiene (**8**) and subjected it to our optimized polymerization conditions (ESI†). Interestingly, the polymerization of **8** required longer reaction times (72 h) to achieve modest conversions (67%), which is likely due to the steric hindrance of incorporating 13-membered rings into the growing polymer chain. The resultant material exhibited poor solubility in common solvents ($CHCl_3$, DCM, THF, PhMe, *etc.*); however, some portions could be dissolved in $CDCl_3$, and subsequent ^{1}H NMR spectroscopic analysis was consistent with the proposed polymer (ESI†). Molecular weight data could only be determined using high-temperature GPC, which revealed that we could access relatively high M_n polymers despite the sluggish nature of the polymerization (M_n = 47 289; $Đ$ = 1.99). Although further work is needed, these preliminary studies demonstrate that lower strain cyclic motifs can successfully be incorporated into polymer matrices.

In conclusion, we have demonstrated that vinyl-addition polymerization of cycloallenes is a potentially valuable strategy

 This journal is © The Royal Society of Chemistry 2020

Published on 26 August 2020. Downloaded by Fudan University on 2/20/2021 3:08:06 AM.

for preparing tunable analogues of cycloolefin polymers. Importantly, cycloallenes can be polymerized in a well-controlled manner at room temperature using a simple Ni catalyst. This route does not require high strain monomers (*e.g.* norbornene) to achieve cyclic motif incorporation, and copolymerization with acyclic monomers is possible (but not required to achieve good conversion). Taken together, these results suggest that vinyl-addition polymerizations of cyclic allenes could provide a reliable synthetic route toward heretofore inaccessible materials.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work was partially supported by the donors of the American Chemical Society Petroleum Research Fund.

Notes and references

1 G. Khanarian, *Polym. Eng. Sci.*, 2000, **40**, 2590–2601.
2 H. Cherdron, M.-J. Brekner and F. Osan, *Die Angew. Makromol. Chem.*, 1994, **223**, 121–133.
3 A.-G. Hoechst, *Ger. Offen*, Germany, 1994, 14. pp.
4 M. O. Liu, H.-F. Lin, M.-C. Yang, M.-J. Lai, C.-C. Chang, M.-C. Feng, P.-L. Shiao and I.-M. Chen, *Polym. Degrad. Stab.*, 2006, **91**, 1443–1447.
5 A. Pegoretti, L. Fambri, C. Lonardi, M. Slouf, A. Dorigato, A. Pegoretti, L. Fambri, C. Lonardi, M. Slouf and J. Kolarik, *eXPRESS Polym. Lett.*, 2011, **5**, 23–37.
6 S. Nishida, H. Nishida and T. Endo, *J. Polym. Sci., Part A: Polym. Chem.*, 2002, **40**, 1151–1156.
7 A. Mizote, T. Tanaka, T. Higashimura and S. Okamura, *J. Polym. Sci., Part A-1: Polym. Chem.*, 1966, **4**, 869–879.
8 A. R. Lavoie and R. M. Waymouth, *Tetrahedron*, 2004, **60**, 7147–7155.
9 S. M. Mitchell, K. A. Niradha Sachinthani, R. Pulukkody and E. B. Pentzer, *ACS Macro Lett.*, 2020, **9**, 1046–1059.
10 T. Yokozawa, N. Ito and T. Endo, *Chem. Lett.*, 1988, **17**, 1955–1958.
11 T. Takahashi, T. Yokozawa and T. Endo, *Macromolecules*, 1995, **28**, 866–869.
12 J. Peska and M. J. Benes, *Collect. Czech. Chem. Commun.*, 1969, **34**, 1115–1117.
13 Y. Takenaka and K. Osakada, *Macromol. Chem. Phys.*, 2001, **202**, 3571–3578.
14 K. Takagi, T. Endo, I. Tomita and T. Endo, *Chem. Commun.*, 1998, 681–682.
15 I. Tomita, Y. Kondo, K. Takagi and T. Endo, *Macromolecules*, 1994, **27**, 4413–4414.
16 M. Taguchi, I. Tomita and T. Endo, *Macromol. Chem. Phys.*, 2000, **201**, 2322–2327.
17 M. Taguchi, I. Tomita and T. Endo, *Angew. Chem., Int. Ed.*, 2000, **39**, 3667–3669.
18 K. Takagi, I. Tomita and T. Endo, *Polym. Bull.*, 2003, **50**, 335–342.
19 K. Takagi, I. Tomita and T. Endo, *Macromolecules*, 1998, **31**, 6741–6747.
20 K. Takagi, I. Tomita and T. Endo, *Chem. Lett.*, 1997, 1187–1188.
21 L. Skattebøl, *J. Org. Chem.*, 1966, **31**, 2789–2794.
22 A. J. Pearson, S. Mallik, A. A. Pinkerton, J. P. Adams and S. Zheng, *J. Org. Chem.*, 1992, **57**, 2910–2917.
23 W. R. Moore and R. C. Bertelson, *J. Org. Chem.*, 1962, **27**, 4182–4186.
24 A series of experiments were conducted to determine if oxidative chain-coupling or catalyst chain transfer could account for the high M_n species observed by GPC (see the ESI).† While these studies failed to elucidate the origin of the high M_n impurity, further investigations are ongoing.
25 C. Liu, J. Yu, X. Sun, J. Zhang and J. He, *Polym. Degrad. Stab.*, 2003, **81**, 197–205.
26 K. J. Daoust, S. M. Hernandez, K. M. Konrad, I. D. Mackie, J. Winstanley and R. P. Johnson, *J. Org. Chem.*, 2006, **71**, 5708–5714.
27 K. R. Gmernicki, E. Hong, C. R. Maroon, S. M. Mahurin, A. P. Sokolov, T. Saito and B. K. Long, *ACS Macro Lett.*, 2016, **5**, 879–883.
28 S. T. Yu, S. J. Na, T. S. Lim and B. Y. Lee, *Macromolecules*, 2010, **43**, 725–730.
29 J. Kolarik, Z. Krulis, M. Slouf and L. Fambri, *Polym. Eng. Sci.*, 2005, **45**, 817–826.
30 B. H. Bowser, C.-H. Ho and S. L. Craig, *Macromolecules*, 2019, **52**, 9032–9038.
31 W. Zhao and K. Nomura, *Macromolecules*, 2016, **49**, 59–70.